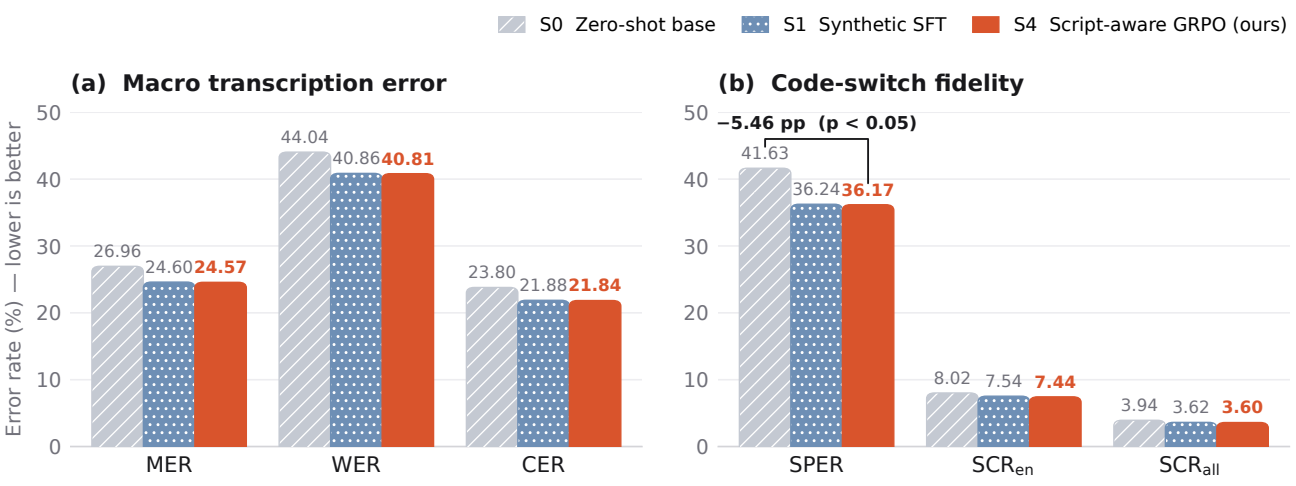

S0 Zero-shot base
S1 Synthetic SFT
S4 Script-aware GRPO (ours)
(a) Macro transcription error
Error rate (%) — lower is better
50
40
30
20
10
0
26.96
24.60
24.57
44.04
40.86
40.81
23.80
21.88
21.84
MER
WER
CER
(b) Code-switch fidelity
−5.46 pp (p < 0.05)
41.63
36.24
36.17
8.02
7.54
7.44
3.94
3.62
3.60
SPER
SCR$_{en}$
SCR$_{all}$